

Computer Networks — Lecture 03

Ch.1 Throughput, Protocol Layers, Security & History — Question Bank with Answers

Part 1: Multiple Choice (25)

1. What is a bottleneck link?

ال bottleneck link ؟

- A. The link with the highest transmission rate on the path
- B. The link that connects the router to the server only
- C. The link on the end-to-end path that constrains the end-to-end throughput
- D. Any link that has no traffic flowing through it

✓ **Correct answer: C. The link on the end-to-end path that constrains the end-to-end throughput**

❓ له صح: ال bottleneck هو الرابط الذي يحدّ من ال throughput الكلي في المسار. ليس الأسرع ولا الفارغ من الحركة.

📖 Ref: Lecture 03 — Section 1.4.4

2. A server is connected to a client through one router. The server-router link rate is R_s and the router-client link rate is R_c . What is the throughput when there is no other traffic?

ب ال R_s والثاني R_c يكون ال throughput ؟

- A. $\min\{R_s, R_c\}$
- B. $R_s + R_c$
- C. $\max\{R_s, R_c\}$
- D. $R_s \times R_c$

✓ **Correct answer: A. $\min\{R_s, R_c\}$**

❓ له صح: ال throughput هو معدل الرابط الأبطأ، أي $\min\{R_s, R_c\}$. الجمع أو الضرب أو الأكبر لا يمثل عنق الزجاجة.

📖 Ref: Lecture 03 — Section 1.4.4

3. For a path with N links of rates $R_1, R_2, \dots, R_N$, the end-to-end throughput is:

فيه N رابط بمعدلات R_1 إلى R_N ، ال throughput الكلي هو:

- A. The average of R_1 to R_N
- B. $R_1 + R_2 + \dots + R_N$
- C. R_N only
- D. $\min\{R_1, R_2, \dots, R_N\}$

✓ **Correct answer: D. $\min\{R_1, R_2, \dots, R_N\}$**

ط الأبطأ في المسار هو من يحدد المعدل الكلي، وليس المتوسط ولا المجموع.

📖 Ref: Lecture 03 — Section 1.4.4

4. According to the lecture, what is typically the constraining factor for throughput in today's Internet?

ال throughput ؟

- A. The Internet core
- B. The access network
- C. The routing protocol
- D. The size of the IP header

✓ **Correct answer: B. The access network**

❓ له صح: قلب الإنترنت (core) مجهّز بروابط عالية السرعة وقليلة الازدحام، فالقيد يكون عادة في شبكة الوصول (access network).

📖 Ref: Lecture 03 — Section 1.4.4

5. In Figure 1.20(b), $R_s = 2 \text{ Mbps}$, $R_c = 1 \text{ Mbps}$, and the common core link $R = 5 \text{ Mbps}$ is shared equally by 10 downloads. What is each download's throughput?

ال 1.20(b): $R_s = 2$ و $R_c = 1$ و $R = 5$ ميغابت/ث تتوزع بالتساوي على 10 تحميلات. كم throughput لكل تحميل؟

- A. 1 Mbps
- B. 2 Mbps
- C. 500 kbps
- D. 5 Mbps

✓ Correct answer: C. 500 kbps

🔗 له صح: الرابط المشترك يعطي كل تحميل $0.5 = 5/10 \text{ Mbps} = 500 \text{ kbps}$ وهو أقل من R_s و R_c فيصبح هو ال bottleneck.

📌 Ref: Lecture 03 — Section 1.4.4

6. In layering, a layer provides its service by:

ال layering، الطبقة تقدّم خدمتها عن طريق:

- A. Performing actions within the layer and using the services of the layer directly below it
- B. Using the services of the layer directly above it only
- C. Bypassing all other layers
- D. Performing actions in every layer of the stack

✓ Correct answer: A. Performing actions within the layer and using the services of the layer directly below it

صح: كل طبقة تنفّذ أعمالاً داخلها وتستخدم خدمات الطبقة التي تحتها مباشرة وتقدّم خدمة للطبقة التي فوقها.

📌 Ref: Lecture 03 — Section 1.5.1

7. Which is a potential drawback of layering mentioned in the lecture?

ال layering؟

- A. It makes hardware more expensive
- B. It prevents the use of TCP
- C. It removes the need for protocols
- D. One layer may duplicate lower-layer functionality

✓ Correct answer: D. One layer may duplicate lower-layer functionality

يُوب تكرار وظائف الطبقات السفلى، وأن طبقة قد تحتاج معلومة موجودة في طبقة أخرى مما يكسر فصل الطبقات.

📌 Ref: Lecture 03 — Section 1.5.1

8. How many layers does the Internet protocol stack have?

ت Internet protocol stack؟

- A. Five
- B. Four
- C. Six
- D. Seven

✓ Correct answer: A. Five

🔗 له صح: المحاضرة: physical و link و network و transport و application أي خمس طبقات. السبع تخص نموذج OSI.

📌 Ref: Lecture 03 — Section 1.5.1

9. Which of the following are application-layer protocols?

أي مما يلي بروتوكولات في طبقة التطبيقات؟

- A. TCP, UDP and IP
- B. HTTP, SMTP and FTP
- C. Ethernet, WiFi and DOCSIS
- D. PPP, IP and UDP

✓ Correct answer: B. HTTP, SMTP and FTP

🔗 له صح: HTTP و SMTP و FTP (و DNS) في طبقة التطبيقات. TCP و UDP و IP و Ethernet و WiFi للرابط.

📌 Ref: Lecture 03 — Section 1.5.1

10. Which service does TCP provide?

هل TCP؟

- A. Connectionless service with no reliability
- B. Routing of datagrams between hosts
- C. Connection-oriented service with guaranteed delivery, flow control and congestion control
- D. Moving individual bits over the medium

✓ **Correct answer: C. Connection-oriented service with guaranteed delivery, flow control and congestion control**

☞ له صح: TCP يقدم خدمة موجهة بالاتصال مع ضمان التسليم والتحكم في التدفق والازدحام. الخدمة بلا اتصال هي UDP.

➦ Ref: Lecture 03 — Section 1.5.1

11. Which statement about UDP is correct?

هل UDP؟

- A. It guarantees delivery of every message
- B. It breaks long messages and throttles the sender
- C. It is a network-layer protocol
- D. It is a no-frills connectionless service with no reliability, flow control or congestion control

✓ **Correct answer: D. It is a no-frills connectionless service with no reliability, flow control or congestion control**

☞ له صح: UDP بسيط وبلا اتصال ولا يضمن شيئاً. ضمان التسليم وتقسيم الرسائل والتحكم في الازدحام من خصائص TCP.

➦ Ref: Lecture 03 — Section 1.5.1

12. Which statement about the IP protocol is correct?

هل IP؟

- A. All Internet components that have a network layer must run the IP protocol
- B. There are several competing IP protocols in the Internet
- C. IP is implemented only in link-layer switches
- D. IP provides reliable end-to-end delivery

✓ **Correct answer: A. All Internet components that have a network layer must run the IP protocol**

☞ له صح: يوجد بروتوكول IP واحد فقط وكل مكون له network layer لازم يشغله، ولذلك يسمى IP layer.

➦ Ref: Lecture 03 — Section 1.5.1

13. How does link-layer reliable delivery differ from TCP's reliable delivery?

هل TCP؟

- A. They are exactly the same service
- B. Link-layer reliability is end-to-end; TCP's is across one link
- C. Link-layer reliability is across one link; TCP's is from one end system to another
- D. Link-layer protocols never provide reliable delivery

✓ **Correct answer: C. Link-layer reliability is across one link; TCP's is from one end system to another**

☞ له صح: بعض بروتوكولات الربط توفر تسليمًا موثوقًا بين عقدتين على رابط واحد، بينما TCP يوفره بين نظامين طرفيين.

➦ Ref: Lecture 03 — Section 1.5.1

14. What is the job of the physical layer?

ما وظيفة الطبقة الفيزيائية؟

- A. Moving entire frames between adjacent elements
- B. Moving the individual bits within a frame from one node to the next
- C. Translating names to IP addresses
- D. Determining routes between hosts

✓ **Correct answer: B. Moving the individual bits within a frame from one node to the next**

لبقة الفيزيائية تنقل البتات الفردية، أما نقل الإطار كاملاً فمهمة طبقة الربط.

➦ Ref: Lecture 03 — Section 1.5.1

15. Which layers does a link-layer switch implement?

ال switch-layer link؟

- A. Layers 1 through 3
- B. Layers 1 through 5
- C. Layers 3 and 4
- D. Layers 1 and 2

✓ **Correct answer: D. Layers 1 and 2**

🔗 له صح: السويتش ينفذ الطبقتين 1 و 2 فقط، والراوتر 1 إلى 3، وال hosts كل الطبقات الخمس.

🔗 Ref: Lecture 03 — Section 1.5.2

16. What does the transport-layer segment consist of?

ال transport-layer segment؟

- A. The application-layer message plus the transport-layer header
- B. The network-layer datagram plus the link-layer header
- C. Only the transport-layer header
- D. The physical-layer bits only

✓ **Correct answer: A. The application-layer message plus the transport-layer header**

🔗 له صح: ال segment يغلف رسالة التطبيق (M) مع رأس النقل (Ht). هذه هي ال encapsulation.

🔗 Ref: Lecture 03 — Section 1.5.2

17. What is the network-layer packet called?

ماذا يسمى حزمة طبقة الشبكة؟

- A. Segment
- B. Datagram
- C. Frame
- D. Message

✓ **Correct answer: B. Datagram**

🔗 له صح: ال datagram و ال segment و ال frame و ال message كلها تسمى حزمة طبقة الشبكة.

🔗 Ref: Lecture 03 — Section 1.5.1 / 1.5.2

18. What is a botnet?

ال botnet؟

- A. A router that blocks malicious traffic
- B. A protocol for encrypting e-mail
- C. A network of thousands of compromised devices controlled by an attacker
- D. A software that records packets passively only

✓ **Correct answer: C. A network of thousands of compromised devices controlled by an attacker**

🔗 له صح: ال botnet شبكة من آلاف الأجهزة المصابة يتحكم بها المهاجم لإرسال spam أو هجمات DDoS.

🔗 Ref: Lecture 03 — Section 1.6

19. In a DDoS attack against a server with access rate R, what must the aggregate attack traffic be approximately?

يتم DDoS على الخادم مع معدل الوصول R، ما يجب أن يكون إجمالي حركة المرور الهجومي تقريباً؟

- A. About R bps
- B. About R / 1000 bps
- C. Exactly zero
- D. Much less than R bps

✓ **Correct answer: A. About R bps**

🔗 له صح: يلزم إجمالي حركة من كل المصادر بحدود R لشل الخدمة. الهجوم الموزع أصعب في الكشف من DoS من مصدر واحد.

🔗 Ref: Lecture 03 — Section 1.6

20. Why are packet sniffers difficult to detect?

الـ packet sniffers ؟

- A. Because they encrypt all packets
- B. Because they run only in wired networks
- C. Because they change the packets' source address
- D. Because they are passive and do not inject packets into the channel

✓ **Correct answer: D. Because they are passive and do not inject packets into the channel**

🔗 له صح: الـ sniffer مستقبل سلبي يسجل نسخًا فقط دون حقن حزم. وأفضل الدفاعات تعتمد على التشفير.

📖 Ref: Lecture 03 — Section 1.6

21. What is IP spoofing?

الـ IP spoofing ؟

- A. Recording copies of all wireless packets
- B. Injecting packets into the Internet with a false source address
- C. Flooding a server with connection requests
- D. Translating host names into IP addresses

✓ **Correct answer: B. Injecting packets into the Internet with a false source address**

🔗 له صح: هو إرسال حزمة بعنوان مصدر مزيف. الحل المذكور end-point authentication. التسجيل هو sniffing.

📖 Ref: Lecture 03 — Section 1.6

22. Why did the Internet become such an insecure place?

لماذا أصبح الإنترنت مكانًا غير آمن؟

- A. Because it was designed without any layers
- B. Because TCP was never part of it
- C. It was originally designed for a group of mutually trusting users on a transparent network
- D. Because it was built only for wireless devices

✓ **Correct answer: C. It was originally designed for a group of mutually trusting users on a transparent network**

لتصميم الأصلي افترض مستخدمين يثق بعضهم ببعض فلا حاجة للأمن، وهذا لم يعد صحيحًا اليوم.

📖 Ref: Lecture 03 — Section 1.6

23. What was the first packet-switched computer network and a direct ancestor of today's Internet?

ما أول شبكة حاسبات بتبديل الحزم وسلف مباشر للإنترنت اليوم؟

- A. ARPAnet
- B. Minitel
- C. Netscape
- D. Napster

✓ **Correct answer: A. ARPAnet**

🔗 له صح: ARPAnet أول شبكة Minitel. Netscape و Napster لمشاركة الملفات.

📖 Ref: Lecture 03 — Section 1.7.1

24. On January 1, 1983, what was officially deployed as the new standard host protocol for ARPAnet, replacing NCP?

في 1 يناير 1983 ما الذي تم نشره رسميًا كبروتوكول قياسي جديد لـ ARPAnet بدل NCP ؟

- A. HTTP
- B. DNS
- C. Ethernet
- D. TCP/IP

✓ **Correct answer: D. TCP/IP**

🔗 له صح: الانتقال من NCP إلى TCP/IP كان flag day أي كل الأجهزة انتقلت في نفس اليوم.

📖 Ref: Lecture 03 — Section 1.7.3

25. Which are the four key components of the Web developed by Berners-Lee and his associates?

ال Web التي طورها Berners-Lee وأتباعه

- A. HTML, HTTP, a Web server and a browser
- B. TCP, UDP, IP and DNS
- C. E-mail, chat, FTP and P2P
- D. Router, switch, hub and modem

✓ **Correct answer: A. HTML, HTTP, a Web server and a browser**

⚡ له صح: المحاضرة تذكر HTML و HTTP وسيرفر ويب ومتصفح. باقي الاختيارات ليست مكونات ال Web.

🔗 Ref: Lecture 03 — Section 1.7.4

Part 2: Essay (10)

1. Explain end-to-end throughput and the bottleneck link for a two-link path and for an N-link path.

ال throughput وال bottleneck link لمسار من رابطتين ولمسار من N رابط.

Model answer:

- Throughput is a critical performance measure besides delay and packet loss.
- Two links (R_s, R_c): throughput = $\min\{R_s, R_c\}$. If $R_c < R_s$ the router backlog grows.
- N links: throughput = $\min\{R_1, R_2, \dots, R_N\}$, the rate of the bottleneck link.
- File transfer time is approximately $F / \min\{\text{rates}\}$.

🔗 Ref: Lecture 03 — Section 1.4.4

2. Why is the access network typically the throughput bottleneck, and how can intervening traffic change the throughput?

ال bottleneck throughput

Model answer:

- The Internet core is over-provisioned with high-speed links that experience little congestion, so throughput = $\min\{R_s, R_c\}$.
- With 10 downloads sharing a core link of rate R, each gets about $R/10$ if the link divides rate equally.
- So a high-rate link can still be the bottleneck if many flows pass through it.

🔗 Ref: Lecture 03 — Section 1.4.4

3. Explain protocol layering: the service model, how layers are implemented, and its drawbacks.

ال layering: ال service model وكيف تُنفَّذ الطبقات وعيوبه.

Model answer:

- Layers give structure to protocol design; each layer offers a service to the layer above (service model).
- Each layer acts within itself and uses the services of the layer below.
- Application and transport layers are in software in end systems; physical and link layers in the network interface card; network layer is mixed.
- Drawbacks: duplicated lower-layer functionality, and a layer may need information from another layer.

🔗 Ref: Lecture 03 — Section 1.5.1

4. Describe the five layers of the Internet protocol stack and the name of the packet at each layer.

ال Internet protocol stack واسم الحزمة في كل طبقة.

Model answer:

- Application: HTTP, SMTP, FTP, DNS; packet = message.
- Transport: TCP and UDP; packet = segment.
- Network: IP and routing protocols; packet = datagram.
- Link: Ethernet, WiFi, DOCSIS, PPP; packet = frame.
- Physical: moves individual bits, depends on the medium (copper, fiber).

🔗 Ref: Lecture 03 — Section 1.5.1

5. Compare TCP and UDP.

بين TCP و UDP.

Model answer:

- TCP is connection-oriented: guaranteed delivery, flow control, breaks long messages into segments, congestion control.
- UDP is connectionless and no-frills: no reliability, no flow control, no congestion control.
- Both transport application-layer messages; the transport packet is called a segment.

🔗 Ref: Lecture 03 — Section 1.5.1

6. Explain encapsulation and which layers hosts, routers and link-layer switches implement.

ال encapsulation وأي طبقات تنفذها ال hosts والراوترات والسويتشات.

Model answer:

- Hosts implement all five layers; routers layers 1 to 3; link-layer switches layers 1 and 2.
- Transport adds header Ht to message M to form a segment.
- Network adds Hn to form a datagram; link adds Hl to form a frame.
- At each layer a packet has header fields and a payload, usually the packet from the layer above.

🔗 Ref: Lecture 03 — Section 1.5.2

7. Describe the three categories of DoS attacks and explain DDoS.

ت DoS الثلاثة و اشرح DDoS.

Model answer:

- DoS makes a network, host or infrastructure unusable by legitimate users.
- Categories: vulnerability attack, bandwidth flooding, connection flooding.
- DDoS: the attacker controls many sources (often a botnet) so aggregate traffic is about R.
- DDoS is much harder to detect and defend against than DoS from a single host.

🔗 Ref: Lecture 03 — Section 1.6

8. Explain malware, botnets, packet sniffing and IP spoofing, and how to defend against them.

ال malware وال botnet وال packet sniffing وال IP spoofing وطرق الدفاع.

Model answer:

- Malware infects hosts, can delete files, install spyware; self-replicating malware spreads exponentially fast.
- Compromised hosts form a botnet used for spam and DDoS.
- A packet sniffer passively records packets and is hard to detect; cryptography is a main defense.
- IP spoofing sends packets with a false source address; end-point authentication is needed.

🔗 Ref: Lecture 03 — Section 1.6

9. Why was the Internet originally insecure?

لماذا كان الإنترنت غير آمن في الأصل؟

Model answer:

- It was designed around a group of mutually trusting users attached to a transparent network, so security was not needed.
- Today's users may not trust each other, may communicate through third parties, and may distrust hardware and software.
- Communication among mutually trusted users is the exception rather than the rule.

🔗 Ref: Lecture 03 — Section 1.6

10. Summarize the main stages in the history of the Internet from 1961 to the new millennium.

من 1961 حتى الألفية الجديدة.

Model answer:

- 1961-1972: packet switching invented; ARPAnet, about 15 nodes by 1972; first e-mail program 1972.
- 1972-1980: TCP, UDP and IP conceptually in place by the end of the 1970s.
- 1980-1990: TCP/IP deployed on January 1, 1983; DNS developed.
- 1990s: the Web (HTML, HTTP, server, browser); killer apps e-mail, Web, instant messaging, P2P.
- 2000s: broadband, video apps, wireless devices passed wired in 2011, social networks, cloud.

🔗 Ref: Lecture 03 — Section 1.7

Part 3: Calculation Problems (5)

11. A server is connected to a client through one router. $R_s = 5$ Mbps, $R_c = 2$ Mbps, and there is no other traffic. Find the throughput and the time to transfer a file of $F = 10$ MB (1 MB = 8 Mb).

تر: $R_s = 5$ و $R_c = 2$ ميغابت/ث ولا توجد حركة أخرى. أوجد ال throughput وزمن نقل ملف 10 ميغابايت.

Model answer:

- Formula: $\text{throughput} = \min\{R_s, R_c\}$; $\text{time} = F / \text{throughput}$.
- $\text{Throughput} = \min\{5, 2\} = 2$ Mbps (bottleneck is the router-client link).
- $F = 10 \text{ MB} \times 8 = 80 \text{ Mb}$.
- $\text{Time} = 80 / 2 = 40$ seconds.

📖 Ref: Lecture 03 — Section 1.4.4

12. A path has 4 links with rates $R_1 = 10$, $R_2 = 4$, $R_3 = 8$ and $R_4 = 6$ Mbps. Find the throughput and the time to send a file of 20 Mb.

من 4 روابط معدلاتها 10 و 4 و 8 و 6 ميغابت/ث. أوجد ال throughput وزمن إرسال ملف 20 ميغابايت.

Model answer:

- Formula: $\text{throughput} = \min\{R_1, \dots, R_N\}$.
- $\text{Throughput} = \min\{10, 4, 8, 6\} = 4$ Mbps (R_2 is the bottleneck).
- $\text{Time} = F / \text{throughput} = 20 / 4 = 5$ seconds.

📖 Ref: Lecture 03 — Section 1.4.4

13. $R_s = 2$ Mbps, $R_c = 1$ Mbps and a common core link $R = 5$ Mbps is shared equally by 10 downloads. Find the throughput of each download.

$R_s = 2$ و $R_c = 1$ و رابط مشترك $R = 5$ ميغابت/ث يتقاسمه 10 تحميلات بالتساوي. أوجد ال throughput كل تحميل.

Model answer:

- Share of the common link = $R / 10 = 5 / 10 = 0.5$ Mbps.
- $\text{Throughput} = \min\{R_s, R_c, R/10\} = \min\{2, 1, 0.5\} = 0.5$ Mbps.
- = 500 kbps. The bottleneck is the shared core link.

📖 Ref: Lecture 03 — Section 1.4.4

14. Same setup with 10 downloads, but $R_s = 3$ Mbps, $R_c = 1.5$ Mbps and the common link $R = 20$ Mbps. Find the throughput of each download and identify the bottleneck.

مع 10 تحميلات لكن $R_s = 3$ و $R_c = 1.5$ و $R = 20$ ميغابت/ث. أوجد ال throughput كل تحميل وال bottleneck.

Model answer:

- Share of the common link = $20 / 10 = 2$ Mbps.
- $\text{Throughput} = \min\{3, 1.5, 2\} = 1.5$ Mbps.
- The bottleneck is the client access link (R_c), not the core.

📖 Ref: Lecture 03 — Section 1.4.4

15. A server has an access rate $R = 100$ Mbps. In a DDoS attack each zombie sends 2 Mbps. Approximately how many zombies are needed to cripple the service?

له $R = 100$ ميغابت/ث. في هجوم DDoS كل zombie يرسل 2 ميغابت/ث. كم zombie

Model answer:

- Principle: aggregate attack traffic must be approximately R .
- Number of zombies = $R / \text{rate per zombie} = 100 / 2 = 50$.
- About 50 zombies are needed.

📖 Ref: Lecture 03 — Section 1.6